



UNIVERSITY OF PERADENIYA
FACULTY OF SCIENCE
SEMESTER II - 2020/2021
MAT1023 - REAL ANALYSIS I
END SEMESTER EXAMINATION

Time allowed: Three Hours

Answer ALL questions

1. (a) State the *Completeness Axiom*.

Determine whether the following sets are bounded (from below, above, or both). If so, determine their infimum and/or supremum and find out whether these infima/suprema are actually minima/maxima.

i. $S_1 = \{1 + (-1)^n : n \in \mathbb{Z}^+ \cup \{0\}\}$	iii. $S_3 = \left\{ \cos\left(\frac{n\pi}{3}\right) : n \in \mathbb{Z}^+ \cup \{0\} \right\}$
ii. $S_2 = \left\{ \frac{1}{m} + \frac{1}{n} : m, n \in \mathbb{Z}^+ \right\}$	iv. $S_4 = \{x \in \mathbb{R} : x^2 + x + 1 \geq 0\}$

- (b) State the *Archimedean Principle*.

- i. State and prove the result for the density of *rational* numbers.
- ii. Using part i., prove the result for the density of *irrational* numbers.

(100 points)

2. (a) Decide which of the following statements are TRUE and which are FALSE. Provide a short proof for the true ones and provide a counterexample with a justification for the false ones.

- i. Every bounded sequence is convergent.
- ii. If $\{x_n\}$ converges and $\{y_n\}$ is bounded, then $\{x_n y_n\}$ converges.
- iii. If

$$x_n = \frac{(n-1) \cos(n^2 + n + 1)}{2n - 1},$$

then $\{x_n\}$ has a convergent subsequence.

- (b) Provide a precise $\varepsilon - N$ argument to prove the existence of $\lim_{n \rightarrow \infty} \frac{2n+3}{3n+4}$.

- (c) State the *Monotone Convergence Theorem* (MCT).

Use the MCT to show that the recursive sequence defined below converges to some number l . What must l be equal to?

$$a_1 = 1, \quad a_{n+1} = \sqrt{2 + a_n}.$$

(100 points)

[Turn over...]

3. (a) Provide a precise $\varepsilon - \delta$ argument to prove that $\lim_{x \rightarrow 0} x^3 \cos(e^{x^2}) = 0$.

(b) State the *Extreme Value Theorem* (EVT).

Provide an example with a justification for each case below to show that dropping the particular assumption leads to the conclusion of the EVT to not to hold.

- i. “closed” assumption of the interval,
- ii. “bounded” assumption of the interval.

(c) If $f : [a, b] \rightarrow \mathbb{R}$ is continuous, prove that $\sup_{x \in [a, b]} |f(x)|$ is finite.

- (d) i. Give an example of functions $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ such that the function defined by $h(x) := f(x) + g(x)$ is continuous in $\mathbb{R}$ but f and g are not continuous at a point.
- ii. Give an example of functions $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ such that the function defined by $h(x) := f(x) + g(x)$ is continuous in $\mathbb{R}$ but f and g are nowhere continuous.

(100 points)

4. (a) Suppose that $f : (0, \infty) \rightarrow \mathbb{R}$ satisfies $f(x) - f(y) = f(x/y)$ for all $x, y \in (0, \infty)$ and $f(1) = 0$.

- i. Prove that f is continuous on $(0, \infty)$ if and only if f is continuous at 1.
(**Hint** : Use the Sequential Characterization of Continuity)
- ii. Prove that f is differentiable on $(0, \infty)$ if and only if f is differentiable at 1.
(**Hint** : Use the definition of derivative of a function)

(b) Verify that the function

$$f(x) = \sqrt{x} - \frac{x}{3}$$

on $[0, 9]$ satisfies all the hypotheses of *Rolle's Theorem*, then find the values of c that satisfy its conclusion.

(c) Show that $f(x) = 2x + e^x$ has exactly one real root.

(**Hint** : Use the *Intermediate Value Theorem* and the *Rolle's Theorem*)

(d) State and prove the *Mean Value Theorem*.

Show that

$$\ln(1+x) < x - \frac{x^2}{2} + \frac{x^3}{3} \text{ for } x > 0.$$

(100 points)

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